CHAPTER 6
QUESTIONS AND ANSWERS

Arduino Microcontroller Concepts (CSE 4820)

1. What is an LDR and how does it work?

Answer: An LDR (Light Dependent Resistor) is a sensor whose resistance changes with light intensity. Its resistance generally decreases in bright light and increases in darkness.

2. Explain how an LDR is interfaced with Arduino using a voltage divider.

Answer: An LDR is connected with a fixed resistor to form a voltage divider. The changing resistance of the LDR produces a changing voltage, which is connected to an Arduino analog input and read using analogRead().

3. Differentiate between digital and analog sensors with examples.

Answer: A digital sensor gives discrete states such as HIGH or LOW. Examples are tilt and motion sensors. An analog sensor produces a continuously varying voltage. Examples are LDR, Piezo and TMP36 sensors.

4. Explain how a tilt sensor can be interfaced with Arduino.

Answer: A tilt sensor can be connected to a digital input pin of Arduino. The sensor behaves like a switch. Arduino reads its state using digitalRead() and can turn an LED or other device ON/OFF according to the detected tilt.

5. Explain how a Piezo sensor can be used for vibration detection.

Answer: A Piezo sensor produces an electrical signal when it experiences vibration or mechanical stress. Arduino reads this signal using analogRead(). The reading is compared with a threshold. If the value exceeds the threshold, vibration is considered detected and an LED or alarm can be activated.

6. Explain how an ultrasonic sensor measures distance using TRIG and ECHO signals.

Answer: Arduino sends a short pulse to the TRIG pin to start the measurement. The sensor sends an ultrasonic wave toward the object. The reflected wave returns to the sensor, and the ECHO signal represents the travel time. Arduino measures this time using pulseIn() and calculates the distance.

7. An ultrasonic sensor measures an echo duration of 2000 µs. Calculate the approximate distance.

Answer: Distance = Time / (29 × 2)

Distance = 2000 / (29 × 2)

Distance ≈ 34.5 cm

Therefore, the object is approximately 34.5 cm away.

8. Explain the voltage-temperature relationship of the TMP36 sensor.

Answer: The TMP36 produces an analog voltage proportional to temperature. It has a 500 mV offset and a sensitivity of 10 mV per °C.

Temperature (°C) = (Vout (mV) − 500) / 10

For example, at 750 mV: Temperature = (750 − 500) / 10 = 25°C.

9. Why is calibration important when interpreting sensor readings?

Answer: Calibration is important because the raw sensor output may not directly represent the actual physical quantity. Calibration helps account for sensor offset, sensitivity, accuracy, noise and environmental effects, allowing the Arduino reading to be converted into a meaningful physical value.

10. Design an Arduino-based system that activates an LED when a sensor value exceeds a selected threshold.

Answer: The system can be designed as:

Sensor → Arduino analog input → analogRead() → Compare with threshold → LED

If the sensor value is greater than or equal to the threshold, the Arduino turns the LED ON. Otherwise, it turns the LED OFF. This technique can be used for light, vibration, sound or temperature detection.